

Advantages of using IPv6 for the smart grid network

Scalability. It provides uniform addressing and seamless connectivity to a larger number of devices.

WAN connectivity. This architecture provides aggregation of many household energy meters/sensors/controllers (typically few hundreds, say 500).

Energy consumption. Each sensor or end node needs to communicate to another device, which is typically located at few meters to few hundred meters; thus the amount of energy consumed will be much lower.

Seamless web services. Applications can be developed independent of the transport network as the applications are not tied to the equipment and can be obtained from the open market.

Open standards. All IP networks are based on true open standards; utilities can mix and match the equipment from multiple vendors across the world.

Simple network architecture. Every device talks IP—no protocol translators, no address translators, no information translators—yielding a simple all-IP network—easy to operate and maintain with low cost of ownership.

End-to-end security. Internet Engineering Task Force (IETF) has developed extensive security protocols for use with IP that provide end-to-end security across multiple heterogeneous interconnected administrative domains.

—Syam Madanapalli, CEO and co-founder, iRam Technologies

Smart meters and harmonics analysis

Harmonics are electrical voltages and currents that appear as a result of non-linear electric loads. "It shows up as an intermediate frequency riding on a fundamental current, which is an interruption to the main current. It is a loss not only for energy but also for motors, transformers and similar equipment," adds Srinivasan.

Harmonics generate a lot of heat as they travel on the surface of the conductor, which affects insulation and drastically reduces equipment life span. A smart grid meter addresses this problem by monitoring the power quality and taking actions in accordance with its observations.

Electromagnetic compatibility and the smart grid

Electromagnetic compatibility (EMC) must be considered to ensure continuous reliable real-time operation in many locations where the smart grid equipment will operate. Components and devices in the smart grid system are subjected to a wide range of conducted and radiated noise sources that are disruptive to all electronic systems (smart grid systems included). These sources can be categorised as follows:

1. Conducted noise from powerline harmonics, surge (from lightning and power system switching transients) and fast transients/bursts (interruption of inductive DC circuits)
2. Radiated noise or signals from known transmitters (AM, FM and TV broadcast transmitters, communications radios, wireless devices, etc.)
3. High power events such as geomagnetic storms, intentional EM interference (IEMI) from portable transmitters and EM pulses associated with high-altitude nuclear detonation (HEMP).
4. Electrostatic discharge (ESD) events when a statically charged body (human or inert) comes in contact with a smart grid device.

—Excerpt from "Electromagnetic compatibility and smart grid interoperability issues" by SGIP Electromagnetic Interoperability Issues Working Group

circuit breaker, DC system and protection circuitry—even whilst the system is on-load. Once the test is completed, it outputs the results on the display.

Multiphysics simulation. The huge engineering project of migrating the electrical grid to a smart grid involves modernising several electrical components. For engineers at ABB Corporate Research Power Technologies in Sweden, multiphysics tools have proved

an invaluable tool for modelling the coupled electromagnetic, thermal and fluid phenomena that take place within these systems.

Data modelling and forecasting

This is where all the data gathered by the sensors, smart meters and other equipment are put into use.

Avoid gold plating the electrical

grid. Unlike how it sounds, 'gold plating' is where everything is over-designed to be safe, by going overboard with thresholds and limitations. Without smart technology to optimise the operation of the power network, gold-plating has been a common approach to ensure that the network is not going to melt during abnormal events.

One of the approaches to overcome gold-plating is the dynamic rating of equipment. This can be applied to overhead lines, underground cables, transformers, etc.

"A recently developed, ground-breaking technology is a 3D model of the electricity network which provides numerous applications including modelling and forecasting applications. An aeroplane loaded with sensors is flown over the network, where it gathers Lidar data point and other relevant data—such as temperature, wind speed, electromagnetic field, heatmaps, etc. An algorithm is run over the Lidar data to automatically extract the utility's assets, resulting into a 3D model and exact position of the utility's assets and nearby environment such as trees," explains Diethelm.

Is an automated power grid secure?

In addition to terrorist activities, natural disasters and ageing equipment can also bring down the electrical grid. Cumulative smart-grid cyber security investment from 2011 to 2018 will total \$14 billion, forecasts Pike Research.

"This requires a better emergency management system with appropriate electronic access control and video surveillance to mitigate problems. The system should also be able to isolate problems and restore electrical services as quickly as possible when disaster happens," says Madanapalli. The existing SCADA systems are more prone to cyber attacks as the utilities start using more and more computer-based applications without an end-to-end design for the smart grid implementation. ●

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